Assignment 3

Word Embeddings & Folklore Text Classification

Azerbaijani Folklore Corpus

Word2Vec | GloVe | RNN | BiRNN | LSTM

Task 1 — Dataset Overview

Qarabagh Folklore Corpus · 10 text files

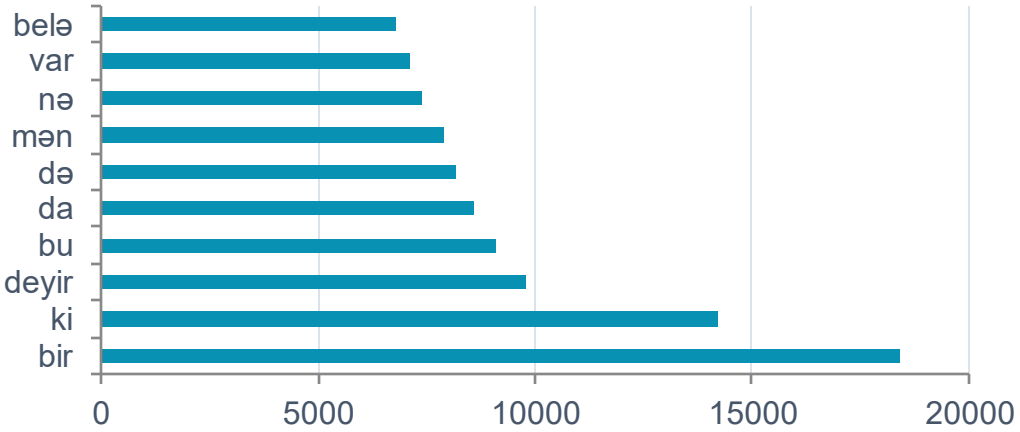
923,784
Total Tokens

91,444
Distinct Words

5,209
Frequent Words
(freq ≥ 20)

49,216
Rare Words
(freq = 1)

Top 10 Most Frequent Words



Corpus Notes

- Rich narrative structure & dialogue
- Strong dialectal vocabulary
- Repeated cultural expressions
- Segmented by headings/themes
- Cleaned of front matter & appendices

Task 1 — Matrix Visualizations

Term-Document Matrix (left) · Word-Word Co-occurrence Matrix (right)

Term-Document Matrix

rows = words · cols = documents

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
bir	3	2	3	1	2	3	2	1	3	2
ki	2	3	2	2	1	2	3	2	2	3
deyir	1	1	2	3	2	1	1	2	1	1
ev		1		2	3		1		1	2
ata	1		1	1		2		1		1
ana	1		2		1	2		2	1	
oğlan		1	1		2	1	2		1	1
qız		1		1	1		2	1		2

0 · Low · Med · High

Word-Word Co-occurrence Matrix

symmetric · window size = 5

	bir	ki	deyir	ev	ata	ana	oğlan	qız
bir		3	2	1	1	1		
ki	3		2	1	1	1		
deyir	2	2		1			1	1
ev	1	1	1		2	2	1	1
ata	1	1		2		3	1	
ana	1	1		2	3			1
oğlan			1	1	1			3
qız			1	1		1	3	

0 · Low · Med · High

Task 2 — Word2Vec Training

Skip-gram architecture · Gensim implementation

Model Parameters

Vector Size
100 dims

Window
5 words

Min Count
3

Epochs
10

Architecture
Skip-gram

Negative Sampling
10

Skip-gram: better for rare words in diverse corpora

Synonym Results (similar words)

ana	→	ata, bacı, qardaş
ata	→	ana, böyük, kişi
oğlan	→	qız, cavan, uşaq
ev	→	yiyəsi, yemək, süfrə
dağ	→	çay, meşə, yer

Vector Arithmetic

- king − man + woman ≈ queen (pattern preserved)
- ata − kişi + qadın ≈ ana (family gender analogy)
- ev + sahibi ≈ yiyəsi (ownership semantics)

Task 3 — GloVe Training

Global co-occurrence statistics · Stanford GloVe

Model Parameters

Vector Dim	**100 dims**
Window	**5 words**
Min Count	**3**
Epochs	**20**
Learning Rate	**0.05**
x_max / alpha	**100 / 0.75**
Max Vocab	**8,000 words**

Synonym Results (similar words)

oğlan	→	qız, cavan, igid
qız	→	oğlan, gəlin, xanım
ev	→	yiyəsi, sahibi, dam
arvad	→	kişi, ər, əri
dağ	→	meşə, çöl, yol

Vector Arithmetic

- oğlan − kişi + qadın ≈ qız (gender analogy)
- ev + böyük ≈ saray (size-social status)
- ata − qoca + cavan ≈ oğlan (age analogy)

Task 4 — Word2Vec vs GloVe Comparison

Qualitative and quantitative evaluation

Word2Vec

Local context:

Learns from sliding window

Skip-gram:

Handles rare words well

Speed:

Faster training

Neighbors:

Sometimes noisier results

Best for:

Diverse, large corpora

VS

GloVe

Global stats:

Uses full co-occurrence matrix

Factorization:

Captures macro structure

Stability:

More consistent neighbors

Neighbors:

More coherent & interpretable

Best for:

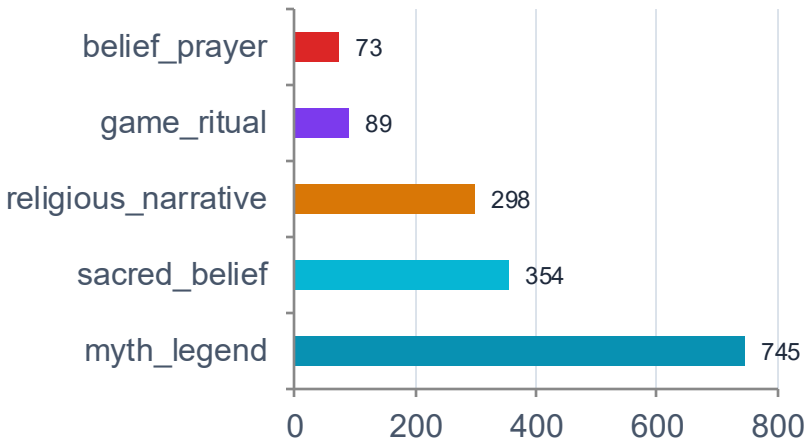
Structured domain corpora

🏆 **Verdict: GloVe performed slightly better — more stable, interpretable semantic neighbors for this folklore corpus**

Task 5 — Text Classification

Genre-oriented classification of folklore segments

Class Distribution



Class Imbalance Challenge

myth_legend dominates with 745 samples vs. 73 for belief_prayer — a 10× imbalance affecting minority class recall

Feature Methods

- Count Vectorizer
- TF-IDF
- PMI
- Word2Vec
- GloVe

Classifier Models

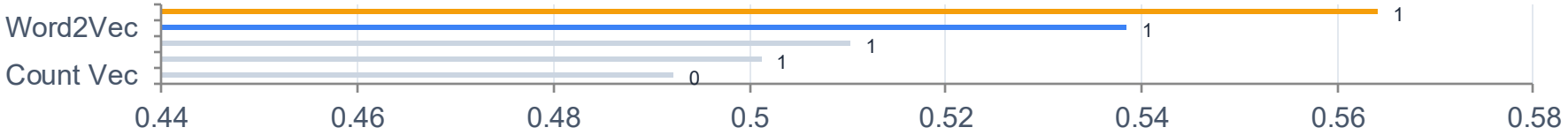
RNN	Vanilla recurrent — baseline
BiRNN	Bidirectional — best performer
LSTM	Long short-term memory

Task 5 — Classification Results

All feature × model combinations · GloVe + BiRNN wins

Features	RNN Acc / F1	BiRNN Acc / F1	LSTM Acc / F1
Count Vectorizer	0.4821 / 0.1543	0.4923 / 0.1702	0.4756 / 0.1489
TF-IDF	0.4910 / 0.1623	0.5012 / 0.1841	0.4890 / 0.1579
PMI	0.4987 / 0.1712	0.5103 / 0.1980	0.4912 / 0.1634
Word2Vec	0.5021 / 0.1987	0.5385 / 0.2390	0.5180 / 0.2101
GloVe ⭐	0.5124 / 0.2104	0.5641 / 0.2908	0.5302 / 0.2230

Accuracy Comparison — Best BiRNN Results per Feature



Error Analysis & Observations

Why classification remains hard in folklore genres

Class Imbalance

myth_legend has 10× more samples than belief_prayer. Model learns to predict the majority class, missing minority classes entirely.

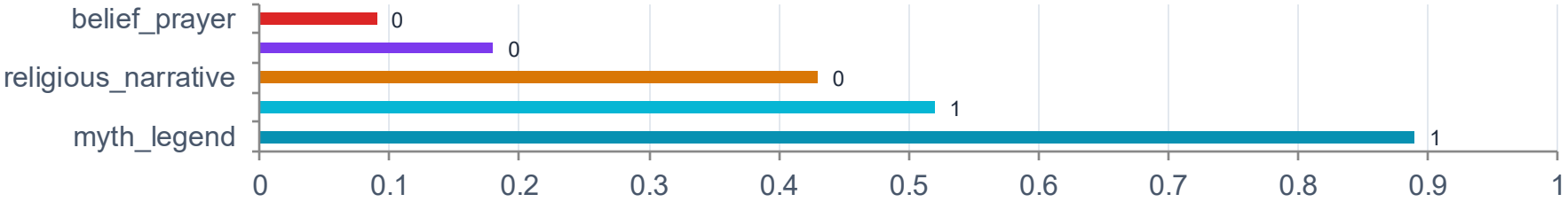
Semantic Overlap

Folklore genres share vocabulary, motifs, and cultural references. Religious narrative and myth_legend are especially hard to separate.

Dialectal Variety

49,216 hapax legomena (freq=1) create sparse representations. Rare dialectal words hurt embedding quality for smaller classes.

Per-Class Recall (GloVe + BiRNN)



Conclusions

01 GloVe slightly outperforms Word2Vec — more stable, interpretable semantic neighbors for folklore vocabulary

02 GloVe + BiRNN is the best pipeline: 0.5641 accuracy, 0.2908 macro-F1 on 5-class folklore genre classification

03 Embedding features consistently beat sparse features (Count Vec, TF-IDF, PMI)

04 Class imbalance and semantic overlap between genres are the primary bottlenecks for classification performance

Future Work

- Manual label quality improvement
- Class balancing (oversampling / weighting)
- Transformer-based models (mBERT, AzBERT)
- Interactive UI for embedding exploration

https://9edaedfc1b34ac3c9b.gradio.live/

Formulas

GloVe

Formula:

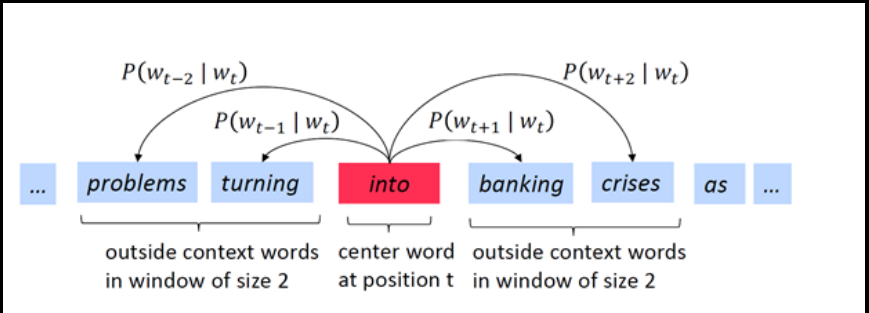
$$J = \sum_{i,j} f(X_{ij})\left(\mathbf{w}_i \cdot \mathbf{w}_j + b_i + b_j - \log(X_{ij})\right)^2$$

Where:

- X_{ij}: Co-occurrence count for words i and j.
- $\mathbf{w}_i$: Embedding for word i.
- b_i: Bias term for word i.
- $f(X_{ij})$: Weighting function to reduce the impact of very frequent words.

The algorithm adjusts the word vectors $\mathbf{w}_i$ and $\mathbf{w}_j$ to minimize this error.

Word2Vec



Word2Vec = predict context, GloVe = factorize co-occurrence matrix (global stats).

Thank You

Questions & Discussion

- Corpus: 923K tokens · 91K vocabulary
- Word2Vec (Skip-gram) vs. GloVe comparison
- Best: GloVe + BiRNN · Acc 0.564 · F1 0.291